Cardinal Stabilization of the Auxiliary Dimension

A substantial partial result for arXiv:2307.08565, Problem 2.8

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Status. Candidate substantial partial result, likely valid, pending human review. The exact dimension for general uncountable index sets remains open.

1 Source problem

Let $\mathcal{H}$ be a nonzero Hilbert space and I a nonempty index set. Dahya asks for the least dimension of a Hilbert space $\mathcal{K}$ such that every commuting family $(S_i)_{i\in I} \subseteq \mathcal{B}(\mathcal{H})$ has a commuting family of C_0-semigroups $(T_i)_{i\in I}$ on $\mathcal{K} \otimes \mathcal{H}$ satisfying

$$T_i(1) = I_{\mathcal{K}} \otimes S_i \qquad (i \in I).$$

The source proves that this least cardinal, denoted here by $\kappa_{\mathcal{H}}(I)$, obeys

$$\aleph_0 \leq \kappa_{\mathcal{H}}(I) \leq \max\{\aleph_0, |I|\},$$

with equality $\aleph_0$ for countable I, and asks whether the upper construction remains optimal for uncountable I [1, Problem 2.8, p. 12].

The embedding problem can also readily be extended to families of operators. In light of out generalisation of the Bhat–Skeide interpolation, we consider the following:

Problem 2.8 Let $\mathcal{H}$ be a Hilbert space and $\mathcal{I}$ a non-empty index set. What is the smallest dimension that an auxiliary Hilbert space $\mathcal{H}'$, can have, such that for each commuting family $\{S_i\}_{i\in\mathcal{I}} \subseteq \mathfrak{L}(\mathcal{H})$ of bounded operators, the commuting family $\{\mathbf{I}_{\mathcal{H}'} \otimes S_i\}_{i\in\mathcal{I}} \subseteq \mathfrak{L}(\mathcal{H}' \otimes \mathcal{H})$ can be *simultaneously embedded* into a commuting family $\{T_i\}_{i\in\mathcal{I}}$ of $\mathcal{C}_0$-semigroups on $\mathcal{H}' \otimes \mathcal{H}$, in the sense that $T_i(1) = \mathbf{I}_{\mathcal{H}'} \otimes S_i$ for $i \in \mathcal{I}$?

Let κ denote this dimension. Lemma 1.4 yields that the κ is bounded above by $\kappa_{\max} := \dim(L^2(\prod_{i\in\mathcal{I}} \mathbb{T}))$. The arguments in the proof of Proposition 2.6 also yield $\kappa \geqslant \aleph_0$. If $\mathcal{I}$ is countable[k] then $\aleph_0 \leqslant \kappa \leqslant \kappa_{\max} = \aleph_0$, whence the choice of auxiliary space in the Bhat–Skeide interpolation is optimal. If $\mathcal{I}$ is uncountable, then $\kappa_{\max} > \aleph_0$.[1] It would be nice to know if our generalisation of the Bhat–Skeide interpolation continues to provide the optimal solution for embedding in this case.

Figure 1: Problem 2.8 and the uncountable-index optimality question in [1, p. 12].

2 Proof intuition

The labels in I carry no information when two labels name the same operator. A family can therefore be compressed to its set of distinct values, which has cardinal at most $|\mathcal{B}(\mathcal{H})|$. Conversely, any compressed family can be padded back to a larger index set by repeating one operator. These two operations prove exact stabilization of the worst-case invariant once the index set is large enough to list every possible operator.

The key point is that repetitions do not require new semigroups: assign the same semigroup to all labels with the same time-one operator. Applying the source paper's construction only after this compression replaces $|I|$ by $\min\{|I|, |\mathcal{B}(\mathcal{H})|\}$. The exact value below that new upper bound is not determined here.

3 Stabilization theorem

Theorem 1 (Index-cardinal stabilization). *Let $\mathcal{H}$ be a nonzero Hilbert space and put $b_{\mathcal{H}} = |\mathcal{B}(\mathcal{H})|$. If I and J are nonempty index sets such that*

$$\min\{|I|, b_{\mathcal{H}}\} = \min\{|J|, b_{\mathcal{H}}\},$$

then

$$\kappa_{\mathcal{H}}(I) = \kappa_{\mathcal{H}}(J).$$

In particular, $\kappa_{\mathcal{H}}(I)$ is constant for all I with $|I| \geq b_{\mathcal{H}}$.

Proof. If $|I|, |J| < b_{\mathcal{H}}$, the displayed hypothesis gives $|I| = |J|$, and a bijection reindexes families and semigroups in both directions.

It remains to treat $|I|, |J| \geq b_{\mathcal{H}}$. Fix an auxiliary Hilbert space $\mathcal{K}$ witnessing the universal property for J, and let $(S_i)_{i \in I}$ be any commuting family on $\mathcal{H}$. Its nonempty set of distinct values

$$A = \{S_i : i \in I\} \subseteq \mathcal{B}(\mathcal{H})$$

has $|A| \leq b_{\mathcal{H}} \leq |J|$. Choose an injection $e : A \to J$ and one $S_0 \in A$. Define a J-indexed family by putting $R_{e(S)} = S$ for $S \in A$ and $R_j = S_0$ on unused coordinates. This family commutes. By the universal property for J, it has commuting C_0-semigroup embeddings $(U_j)_{j \in J}$ on $\mathcal{K} \otimes \mathcal{H}$.

For each $i \in I$, set $T_i = U_{e(S_i)}$. The family $(T_i)_{i \in I}$ commutes and

$$T_i(1) = U_{e(S_i)}(1) = I_{\mathcal{K}} \otimes R_{e(S_i)} = I_{\mathcal{K}} \otimes S_i.$$

Thus every auxiliary dimension valid for J is valid for I, so $\kappa_{\mathcal{H}}(I) \leq \kappa_{\mathcal{H}}(J)$. Interchanging I and J proves the reverse inequality. □

Theorem 2 (Explicit operator cardinality). *If $\delta = \dim \mathcal{H}$ and $\theta = \max\{\delta, \aleph_0\}$, then*

$$|\mathcal{B}(\mathcal{H})| = 2^{\theta}.$$

Proof. Scalar multiples of the identity give $2^{\aleph_0}$ distinct operators. If $(e_\alpha)_{\alpha < \delta}$ is an orthonormal basis, the orthogonal projections onto the closed spans of arbitrary subsets of this basis give 2^{δ} distinct operators. Hence $|\mathcal{B}(\mathcal{H})| \geq 2^{\theta}$.

For the reverse inequality, every vector has countable support in an orthonormal basis. There are at most 2^{θ} countable supports and at most 2^{θ} coefficient sequences, so $|\mathcal{H}| \leq 2^{\theta}$. An operator is determined by its values on the δ basis vectors. Therefore

$$|\mathcal{B}(\mathcal{H})| \leq |\mathcal{H}|^{\delta} \leq (2^{\theta})^{\delta} = 2^{\theta},$$

where the finite-dimensional case is the same elementary continuum count. □

Corollary 3 (Improved upper bound and nonoptimality). *For every nonempty I,*

$$\aleph_0 \leq \kappa_{\mathcal{H}}(I) \leq \max\Big\{\aleph_0, \min\Big(|I|, 2^{\max\{\dim \mathcal{H}, \aleph_0\}}\Big)\Big\}.$$

If $|I| > |\mathcal{B}(\mathcal{H})|$, the source construction of auxiliary dimension $|I|$ is not optimal. In particular, for nonzero separable $\mathcal{H}$ and $|I| > 2^{\aleph_0}$,

$$\kappa_{\mathcal{H}}(I) \leq 2^{\aleph_0} < |I|.$$

Proof. Choose J with $|J| = \min\{|I|, b_{\mathcal{H}}\}$. Theorem 1 gives $\kappa_{\mathcal{H}}(I) = \kappa_{\mathcal{H}}(J)$. Lemma 1.4 of the source and its cardinal calculation give $\kappa_{\mathcal{H}}(J) \leq \max\{\aleph_0, |J|\}$ [1, Lemma 1.4 and Problem 2.8]. The source lower bound is $\aleph_0$. Now apply Theorem 2. $\square$

Remark 4. *The threshold $b_{\mathcal{H}}$ is not inflated by allowing noncommuting operators. The scalar family has cardinal $b_{\mathcal{H}}$ in finite dimension, while the diagonal projection family has cardinal $b_{\mathcal{H}}$ in infinite dimension. Thus a commuting family can have the maximum possible number of distinct operator values.*

4 What remains open

The result gives an exact stabilization law and a strict improvement when $|I| > |\mathcal{B}(\mathcal{H})|$, but it does not compute the stabilized value $\kappa_{\mathcal{H}}(b_{\mathcal{H}})$. It also does not decide optimality when $\aleph_0 < |I| \leq |\mathcal{B}(\mathcal{H})|$.

Eight focused routes were audited. Compression by distinct values and the exact cardinality computation succeeded. Replacing $\mathcal{B}(\mathcal{H})$ by a bound for commuting sets gives no improvement, because diagonal families attain the same cardinal. A common clock works for some projection families but the wrap cocycles in the Bhat–Skeide construction do not commute for arbitrary operators. Algebra-generator compression does not canonically interpolate all functions of the generators. Counting strongly continuous semigroups produces no auxiliary lower bound because the original $\mathcal{H}$ factor already stores $2^{\dim \mathcal{H}}$ possibilities. Joint coordinate shifts likewise put the large cardinality into $\mathcal{H}$, not the auxiliary factor. No credible remaining route produced either a universal smaller construction or a lower bound above $\aleph_0$.

5 Verification and bounded novelty search

The proof is symbolic. The principal checks are: (i) the padded J-family is commuting; (ii) assigning one semigroup to every repeated value preserves pairwise commutation and the time-one identities; (iii) both inequalities in Theorem 1 are obtained symmetrically; and (iv) the cardinal computation treats finite and infinite Hilbert dimension.

The run's registry, solution, attempt, and proof-gap indexes were searched by exact arXiv id and the core phrases "Problem 2.8", "auxiliary Hilbert space", "simultaneously embedded", and "Bhat–Skeide". A bounded exact-title and exact-phrase search of arXiv and the publisher page through 17 August 2026 found the source paper and citations but no later resolution or exact stabilization statement. This is bounded evidence, not a priority claim.

Human-review recommendation. Promote for expert review as a substantial partial result. Check the interpretation of the universal least dimension, the reindexing construction in both directions, and whether the same stabilization observation appears in later work under different terminology.

References

[1] R. Dahya, *Interpolation and non-dilatable families of $\mathcal{C}_0$-semigroups*, arXiv:2307.08565 (2023; revised 2024), published in Banach Journal of Mathematical Analysis 18 (2024), article 34.